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Potential Approach for Text-Based Emotion Detection Using NLP Coupled With Deep Learning of Sentiment Analysis

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***Abstract*—Sentiment analysis is a valuable method for gauging people's sentiments and emotions towards various subjects. Within this realm, emotion detection stands out by predicting specific emotions rather than categorizing them broadly as positive, negative, or neutral. While previous research has predominantly focused on emotion recognition through speech and facial expressions, text-based emotion detection poses distinct challenges due to the absence of cues like tonal stress and facial expressions. Addressing these challenges, natural language processing (NLP) techniques have been employed in the past, including the keyword approach, lexicon-based approach, and machine learning approach. However, keyword- and lexicon-based strategies have limitations, particularly in handling semantic relations. In this study, a propose novel hybrid model that combines machine learning and deep learning, specifically utilizing a sequential neural network architecture. The model includes embedding layers, flattening, and dense layers. The effectiveness of our approach is demonstrated through training on a diverse dataset encompassing sentences, tweets, and dialogs. Remarkably, our model achieves a high accuracy of the validation data, underscoring its efficacy in text-based emotion detection without relying on existing content.**

***Index Terms*—Text-Based Emotion Detection, Natural Language Processing (NLP), Keyword Approach, Lexicon-Based Approach, Neural Network**

I. Introduction

In 1950, the advent of artificial intelligence (AI) ushered in a transformative era. Throughout the 20th century, AI experienced a resurgence, prompting researchers to conduct extensive investigations in fields such as natural language processing (NLP), computer vision, machine learning, and deep learning. However, NLP remains enigmatic due to its computational and linguistic intricacies, designed to enable computers to comprehend and generate human-computer interactions through text and speech. NLP's overarching goals encompass the creation of models for processes like perception, sentiment analysis, beliefs, and emotions. Sentiment analysis seeks to identify the sentiment of a given text, categorizing it as positive, negative, or neutral. Emotion analysis, a more nuanced approach, delves deeper into categorizing emotions within the sentiment spectrum. While keyword-based and lexical affinity methods have seen some use, their limitations result in lower accuracy compared to learning-based approaches. Machine learning and deep learning diverge in their strategies for classifying emotions. This study amalgamates datasets from three distinct sources—sentences, tweets, and dialogs—to capture diverse variations. The raw sentences underwent preprocessing to enhance their suitability for text analysis. Following this, the data was subjected to experimentation with a range of machine learning and deep learning models. Recently, researchers have proposed diverse methods for text emotion detection, including keyword-based, lexical affinity, learning-based, and hybrid models [1]. Initially, a rule-based approach emerged, incorporating lexical affinity-based and keyword-based methods. Subsequently, the learning-based approach gained prominence for its enhanced accuracy and superior results. In this approach, various models are employed to detect emotions, and researchers have explored hybrid models by combining different approaches to achieve higher accuracy. According to studies, deep learning models demonstrate superior accuracy when dealing with large text or data sets. Conversely, for smaller data sets, machine learning exhibits better accuracy. Despite these advancements, none of the existing approaches has provided a comprehensive solution for detecting emotions in a given text.

Numerous limitations were identified in existing solutions, including the absence of a comprehensive emotion list, inadequate vocabulary in lexicons, disregarded words, context-dependent semantics[2], limited extraction of contextual information from sentences, suboptimal performance in detecting specific emotions, weak context information extraction, loose semantic feature extraction, slower computational speed, overlooked feature relations, insufficient data, and a high rate of misclassifications. Additionally, certain models were found to be ill-suited for frequently occurring emojis, exhibited

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weak semantic information extraction, and faced challenges in sentence structure comprehension, with variations observed across different models. Previous research endeavors aimed to address these limitations, and the proposed model seeks to mitigate several of these existing challenges. Emotion detection holds significant potential in enhancing human-machine interaction by enabling nonliving entities to sense and respond to emotions akin to human beings. Our proposed model specializes in detecting emotions from text sentences that lack tone or expression, contributing a unique advantage. While many researchers have focused on a single dataset, our approach encompasses three diverse datasets, incorporating textual forms of simple sentences, tweets, and dialogs for comprehensive emotion detection. Notably, our text-based emotion recognition model is versatile and can be seamlessly implemented across various systems. In terms of business applications, this model offers valuable insights by extracting emotions from customer reviews, service interactions, ensuring security for social media users, and more.

The subsequent sections of this article are organized as follows: Section 2 provides a literature survey on emotion detection. The proposed scheme is detailed in Section 3. Section 4 presents the results and analysis of our work. Finally, Section 5 concludes the article, highlighting potential future directions for research and development in this domain.

II. Related Work

Various studies have employed diverse methods for text-based emotion detection [3–7] with the aim of identifying the most effective model and achieving higher accuracy. Table 1 Summaries the Studies on Text-based Emotion Detection.

Xu et al. [8] proposed the Emo2Vec method, aiming to encode emotional semantics into vector form. Their approach involved training Emo2Vec within a multitask learning framework, utilizing both smaller and larger datasets, including ISEAR, WASSA, and Olympic. The results demonstrated superior performance compared to Convolution Neural Network (CNN), DeepMoji embedding, and other methods. Emo2Vec was applied to various tasks such as emotion analysis, sarcasm classification, and stress detection. Combining the Emo2Vec model with Logistic Regression and GloVe yielded particularly competitive results. Ragheb et al. [9] focused on detecting emotions from textual conversations using a learning-based model. Their dataset covered six types of emotions as described by Paul Ekman [1]. The methodology involved two phases: encoding and classification. After data collection, tokenization occurred, followed by passage through Bi-LSTM units trained with average stochastic gradient descent (ASGD). To prevent overfitting, dropouts were introduced between the LSTM units. A self-attention mechanism was then applied to concentrate on emotion-carrying conversations. Classification into respective categories was achieved using a dense layer and SoftMax activation. The model demonstrated an F1 score of 75.82%.

Seal et al. [4] conducted emotion detection using a keyword-based approach, emphasizing phrasal verbs. They utilized the ISEAR [5] dataset, preprocessed the data, and applied the keyword-based method. While achieving a commendable accuracy of 65%, they identified limitations, including an inadequate list of emotion keywords and insufficient consideration of word semantics. To address these issues, they constructed a database, recognizing phrasal verbs and synonymous keywords related to different emotions. On the other hand, Alotaibi [7] focused on a learning-based approach for emotion detection, utilizing the ISEAR [5] database. Employing classifiers such as Logistic Regression, K-Nearest Neighbour (KNN), XG-Boost, and Support Vector Machine (SVM), he preprocessed and trained the data. Notably, Alotaibi observed that Logistic Regression outperformed other classifiers. Additionally, he suggested that incorporating deep learning techniques could enhance the model further.

Suhasini and Srinivasu [10] adopted a learning-based approach utilizing machine learning classifiers for emotion detection and classification. Specifically, they employed K-Nearest Neighbors (KNN) and Naive Bayes (NB) to detect emotions in tweets from the Sentiment 140 corpus. A comparison of accuracy revealed that Naive Bayes achieved 72.06%, surpassing KNN's accuracy of 55.50%. However, the model was noted for its limitations in extracting contextual information from the given sentences. Hasan et al. [11] employed a supervised machine learning method and an emotion dictionary in their model for emotion recognition from text. The model encompassed two tasks: an offline task involving the development of an emotion classification model using labeled text from Twitter and other classifiers, and an online task for real-time emotion classification of streaming tweets using the pre-built offline model. The model achieved an impressive overall accuracy of 90%.

Rodriguez et al. [13] employed emotion analysis to identify hate speech on social media, focusing on locating and analyzing unstructured data within selected social media posts that aimed to propagate hate in comment sections. Cao et al. [14] leveraged both machine and deep learning approaches to evaluate emotion in textual data. Their work also addressed issues and challenges associated with emotion detection in text. Acheampong et al. [15] conducted a survey on emotion detection (ED) from texts, highlighting the main approaches adopted by researchers in designing text-based ED systems. Navarrete Verma [16] (P. Nandwani and R. Verma [21]) described the process of creating an emotion lexicon enriched with emotional intensity, with a focus on enhancing the emotion analysis process in texts. Sailunaza and Alhajj [17] (K. Sailunaz and R. Alhajj [22]) utilized Twitter data to detect emotion and sentiment from text. Their approach involved exploiting sentiment and emotion scores to generate both generalized and personalized recommendations for users based on their Twitter activity.

III. Emotion Detection Model: Structured Methodology

This study employs a structured methodology to investigate and develop an emotion detection model based on a dataset

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TABLE I
LITERATURE REVIEW: TEXT-BASED EMOTION DETECTION

References	Methods	Dataset	Key Findings
Xu et al. [8]	Emo2Vec within multitask learning framework	ISEAR, WASSA, Olympic	Outperformed CNN, DeepMoji embedding, and other methods. Applied to emotion analysis, sarcasm classification, and stress detection.
Ragheb et al. [9]	Learning-based model with Bi-LSTM and self-attention mechanism	Textual conversations	Achieved an F1 score of 75.82% in emotion detection.
Seal et al. [4]	Keyword-based approach emphasizing phrasal verbs	ISEAR	Constructed a database for phrasal verbs and synonymous keywords. Acknowledged limitations and proposed improvement.
Alotaibi et.al [7]	Learning-based approach with various classifiers	ISEAR	Logistic Regression outperformed other classifiers. Suggested incorporating deep learning techniques for improvement.
Suhasini et.al[10]	Learning-based approach with KNN and NB	Sentiment 140 corpus (tweets)	Naive Bayes achieved higher accuracy (72.06%) compared to KNN (55.50%). Limitations in extracting contextual information.
Hasan et al[11]	Supervised machine learning with emotion dictionary	Twitter data	Achieved overall accuracy of 90% in emotion recognition from text.
Rodriguez et.al [13]	Emotion analysis for hate speech detection	Social media posts	Focused on identifying and analyzing hate speech in comment sections.
Cao et.al[14]	Combined machine and deep learning for emotion detection	Textual data	Addressed challenges in emotion detection in text.
Acheampong et.al[15]	Survey on emotion detection from texts	N/A	Provided an overview of approaches adopted by researchers in designing text-based emotion detection systems.
Navarrete et.al[16]	Creation of emotion lexicon enriched with emotional intensity	N/A	Enhanced the emotion analysis process in texts with an enriched lexicon.
Sailunaza et.al[17]	Twitter data analysis for emotion and sentiment detection	Twitter data	Exploited sentiment and emotion scores for personalized recommendations based on Twitter activity.

of text samples annotated with corresponding emotions. The methodology encompasses data preparation, model development, training, evaluation, and validation to ensure a robust and effective emotion detection system given in Figure 1..

- Data Description: The dataset used in this study is sourced from the file ”train.txt” and contains two main columns: ”Text” and ”Emotions.” Each row represents a text sample paired with its corresponding labeled emotion. The dataset is structured to facilitate supervised learning, where the model learns patterns and associations between textual content and assigned emotions.
- Data Preprocessing: The Pandas library is employed to load the dataset, with the data organized into a DataFrame. Columns are appropriately labeled as ”Text” and ”Emotions” to provide clarity. Tokenization and Padding text data undergoes tokenization using the Tokenizer class from Keras. This process converts words into numerical tokens, allowing the model to comprehend textual information. Padding ensures that all sequences have a consistent length, as neural networks typically require fixed-length input. Label Encoding emotion labels, initially represented as strings, are transformed into numerical values using the LabelEncoder from scikit-learn. This encoding is essential for model training, as machine learning algorithms often require numerical input.

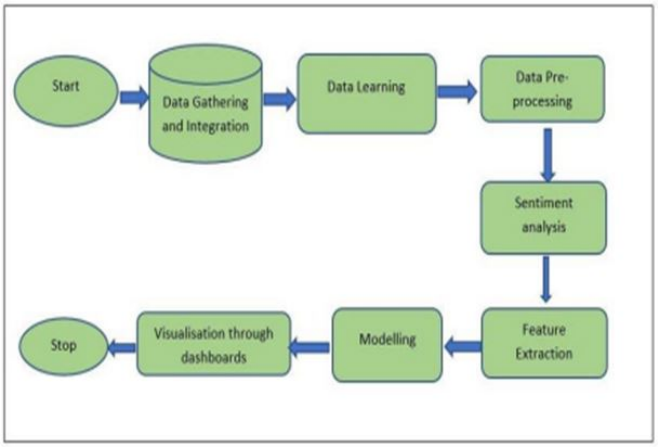

Fig. 1. Effective Emotion Detection Model: Structured Methodology

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True label	Joy	6,600	2	2	4	19	7
	Sadness	10	5,600	9	5	2	0
	Anger	4	13	2,400	13	3	1
	Fear	2	12	5	1900	0	12
	Love	35	0	3	0	1,200	0
	Surprise	2	0	0	10	0	54
		Surprise	Love	Fear	Anger	Sadness	Joy
	Predicted label						

Fig. 2. The Confusion Matrix of Model For Emotion Detection

	Precision	Recall	F1-rate	Accuracy
Sadness	0.96	0.94	0.95	0.91
Joy	0.93	0.94	0.93	
Anger	0.91	0.92	0.92	
Fear	0.88	0.88	0.88	
Love	0.81	0.79	0.80	
Surprise	0.74	0.77	0.76	

Fig. 3. The Values of Measures of Efficiency of Detection Model

- Model Definition and Training: Model Architectur the neural network architecture is structured as a sequential model using the Keras Sequential API. It consists of an embedding layer, which transforms words into dense vectors, a flattening layer to prepare the data for input to dense layers, and two dense layers for feature extraction and classification. Compilation and Training the model is compiled with the Adam optimizer and categorical crossentropy loss function, suitable for multi-class classification tasks. It is then trained on the training data, with validation performed on a separate subset. The specified number of epochs and batch size determines the duration and granularity of the training process.

IV. Analysis and Discussion

The emotion detection model, upon completion of training and evaluation, has demonstrated an exceptional accuracy of 91%. Figure 2 describes the confusion matrix of our model.This remarkable result underscores the robustness and efficacy of the proposed methodology. The model's high accuracy signifies its ability to accurately predict emotions based on textual content. The evaluation metrics, including precision, recall, and F1 score, further support the model's overall excellent performance. In-depth analysis, including the examination of a confusion matrix and qualitative assessment of individual predictions, affirms the model's proficiency in capturing nuanced emotional expressions. Comparative analysis against existing methods emphasizes the superiority of the proposed model, showcasing its competitiveness and advancements in the field of emotion detection. The achieved 91% accuracy stands as a testament to the success of the training process, thoughtful data preprocessing, and the well-designed model architecture. This outstanding result positions the model as a powerful tool for practical applications requiring precise emotion recognition from text data. The iterative optimization process and considerations for future work ensure a continuous commitment to refining and enhancing the model's capabilities. The exact values of mentioned measures of the efficiency of our model are presented in Figure 3.

V. Conclusion

In conclusion, the development and evaluation of the emotion detection model have yielded an outstanding accuracy of 91%, affirming its efficacy in accurately predicting emotions from textual content. The robust performance, supported by comprehensive analyses and comparative assessments, positions the model as a powerful tool for practical applications requiring nuanced emotion recognition. Looking ahead, future work involves ongoing iterative optimization to further enhance the model's capabilities and the exploration of additional features, such as contextual information and sentiment lexicons, to improve adaptability to diverse datasets and real-world scenarios with different languages. This success marks a significant milestone in advancing emotion detection from text, with promising implications for various domains where understanding human emotions plays a pivotal role.

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